its length and corresponding change in diameter. Therefore, the resistance of the wire is changed. This change in resistance is measured in terms of applied pressure.

e. Bonded strain gauge

These gauges are directly bonded (that is pasted) on the surface of the structure under study. Hence, they are termed as bonded strain gauges. Bonded strain gauges offer a rugged assembly and good accuracy that is not degraded by shock and vibration. However, bonded strain gauges are limited in their pressure and temperature ranges. The three types of bonded strain gauges are

i. Fine wire strain gauge
ii. Metal foil strain gauge
iii. Semi-conductor gauge

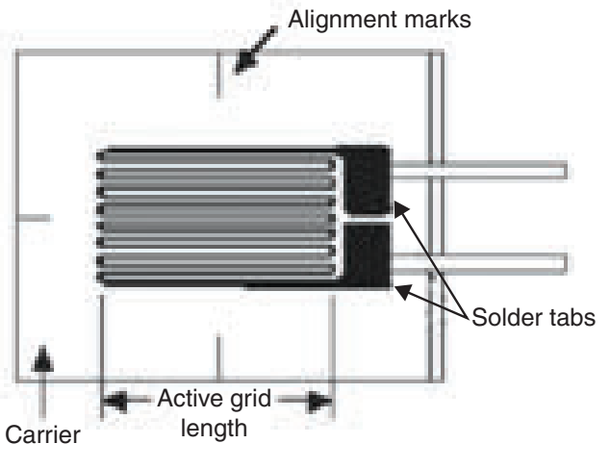


Fig. 7.6 Bonded strain gauge

f. Resistance thermometers

The resistance of a conductor changes when its temperature is changed. This property is utilized for measurement of temperature. The resistance thermometer detector (RTD) uses the change in electrical resistance of conductor to determine the temperature. Platinum and nickel are used for the measurement of temperature being it can withstand high temperature. These types of temperature detector are having positive temperature coefficient.

The platinum element is placed inside the RTD and connecting wires are pulled out from the RTD to the external circuit. The external circuit may be wheatstone bridge. The temperature variation causes the deflection in the bridge which is the measure of temperature. The mounting threads are used to mount this detector to the equipment of temperature measurement.



Fig. 7.7 Industrial platinum RTD

g. Thermistor

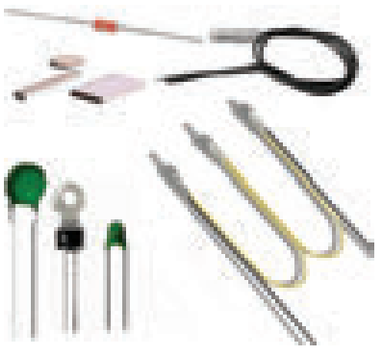


Fig. 7.8 Thermistor

Another type of resistive transducer is thermistor whose resistance depends on temperature. Thermistors are available in the form of beads, rods and discs.

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